

Assignment-1

Due on 02/09/2026

Note-1: Please submit the Programming and Non-programming (*handwritten and scanned*) component as one compressed file only.

Note-2: Students may be called for a viva voce to assess their understanding of the concepts, methodology, implementation, and results presented in the submitted assignment.

Non-programming:

[1] *The RL Bookshelf Challenge*: List the important classical textbooks in RL and, for each book, briefly highlight its distinctive treatment and emphasis. What makes each book different from the others

[2] Solve the following problems from the textbook "Sutton, R. S., & Barto, A. G. (2018). Reinforcement learning: An introduction (2nd ed.). MIT Press." link: <http://incompleteideas.net/book/the-book-2nd.html>

- Exercise 3.1
- Exercise 3.2
- Exercise 3.3
- Exercise 3.7
- Exercise 3.8
- Exercise 3.12
- Exercise 3.14
- Exercise 3.22

[3] Suppose we want an agent to learn to play the game of golf. For training purposes, we make use of a golf simulator program. Assume that the original reward distribution gives a reward of +10 when the golf ball is hit into the hole and -1 for all other transitions. To aid the agents learning process, we propose to give an additional reward of +3 whenever the ball is within a 1-metre radius of the hole. Is this additional reward a good idea or not? Why?

Programming:

[4] **MDP in Python and basics of Pytorch**

Review the problem statements provided in the notebook [skeleton-code](#) and complete all the tasks given.